

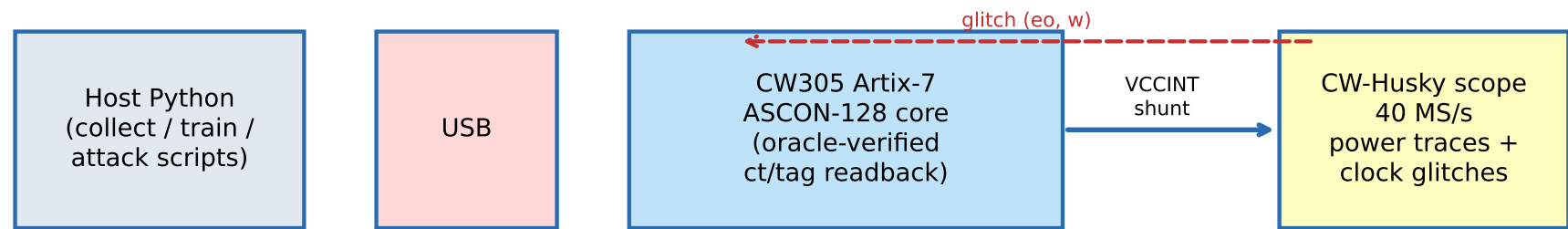
CW305-Based Physical Security Evaluation of ASCON-128:

Deep-Learning Side-Channel Analysis and Clock-Glitch Fault Injection

Prajwal R • Prajwal G S Basavaraj • Sachin S • Nitesh
Dept. of Electronics and Communication Engineering, PES University, Bengaluru, India
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Motivation & Platform

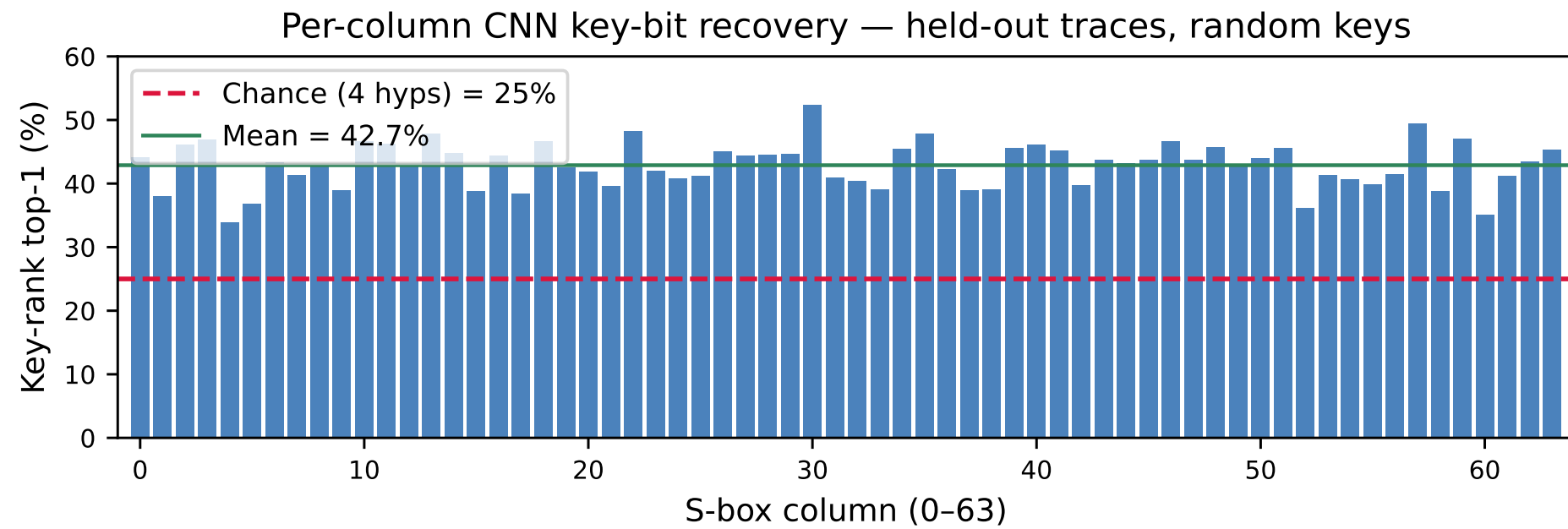
ASCON-128, the NIST SP 800-232 lightweight AEAD standard, is entering constrained FPGA deployments, yet hardware-measured leakage and fault data on the official unmasked NIST LWC core remain sparse. We ask: what does a real Artix-7 implementation leak under power analysis and clock glitching — and which apparent results are measurement artifacts?



- CW305 Artix-7 + CW-Husky scope; 40 MS/s against 10 MHz crypto clock (4 samples/cycle), VCCINT shunt, per-query trigger on `tio4`.
- Byte-exact oracle verification: every stored trace is accepted only after its ct/tag readback matches a software reference (5/5 NIST KATs on the bitstream).
- Random keys for profiling; nonce-repetition averaging M as an explicit SNR knob ($\sqrt{M}$ gain); fixed AD/PT blocks.

Deep-Learning SCA: the S-box Target

One CNN/MLP profile per S-box column (64 columns $\times$ 2 unknown key bits $\Rightarrow$ 4 hypotheses/column), trained on 80/20 random-key splits; HW labels 0-5; empty-class-weighted loss.

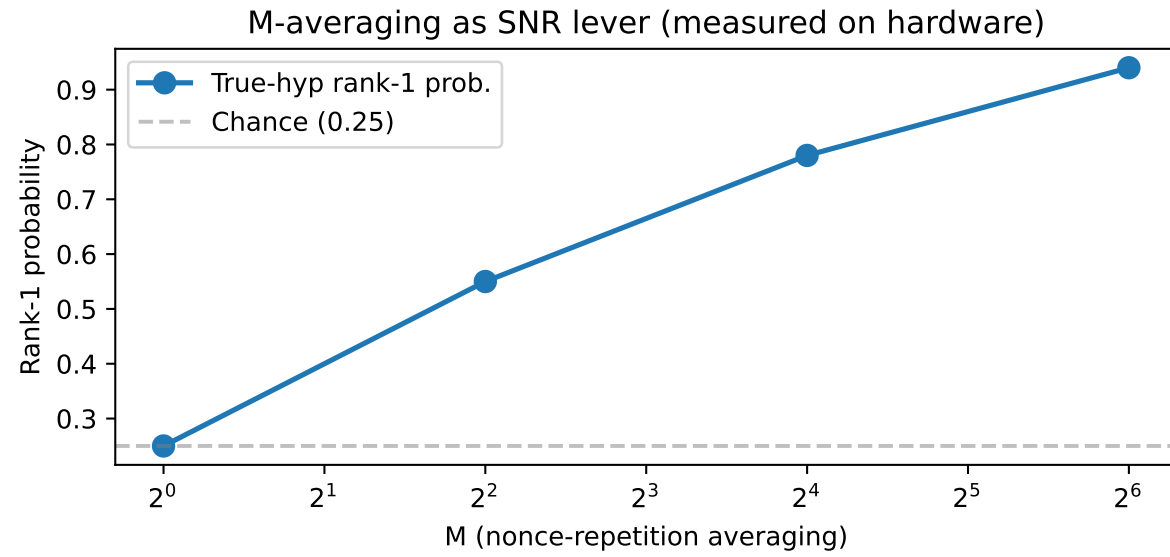


- Per-column key-bit recovery on held-out random-key traces: key-rank top-1 42.7% mean vs 25% chance; all 64 columns above chance, 0/64 below 30% (offline-verified; HW-class top-1 28-35% vs 16.7% uniform chance).
- Profiling-domain gap is real and column-specific: offline profiles transfer poorly to the live board across sessions.
- Live on-board fine-tuning closes the gap: 300 board traces at a known key $\Rightarrow$ 90.7% training accuracy vs 72% majority floor; the fine-tuned column recovers correct bits on the live board at $M=1$, while unfine-tuned columns sit at chance.

Full-Key Assembly Strategy

Divide-and-conquer: (1) closed-loop attack per column $\Rightarrow$ posterior $P(h_c | t)$ over 4 hypotheses; (2) top-1 per column $\Rightarrow$ 128-bit candidate; (3) verify by re-encrypting a fresh nonce against the oracle; (4) bounded enumeration over the least-confident columns (2^n). Budget $\approx 64 \times q(M) \times 2^n$. Characterized as ongoing work: the empirical 64-column run needs per-column on-board fine-tuning — a measurement-budget constraint, not an information-theoretic one.

The SNR Lever: Nonce-Repetition Averaging



- Repeating one separating nonce M times and averaging raises per-query SNR by $\sqrt{M}$.
- True-hypothesis rank-1: 0.25 at $M=1$ (chance) $\rightarrow$ 0.94 at $M=64$ (on-board probe, one column).
- In the closed loop, $M=64$ shortens convergence 15 $\rightarrow$ 9 queries/column.
- The align-then-z-score pipeline removes jitter and DC drift so the full $\sqrt{M}$ gain survives end-to-end.

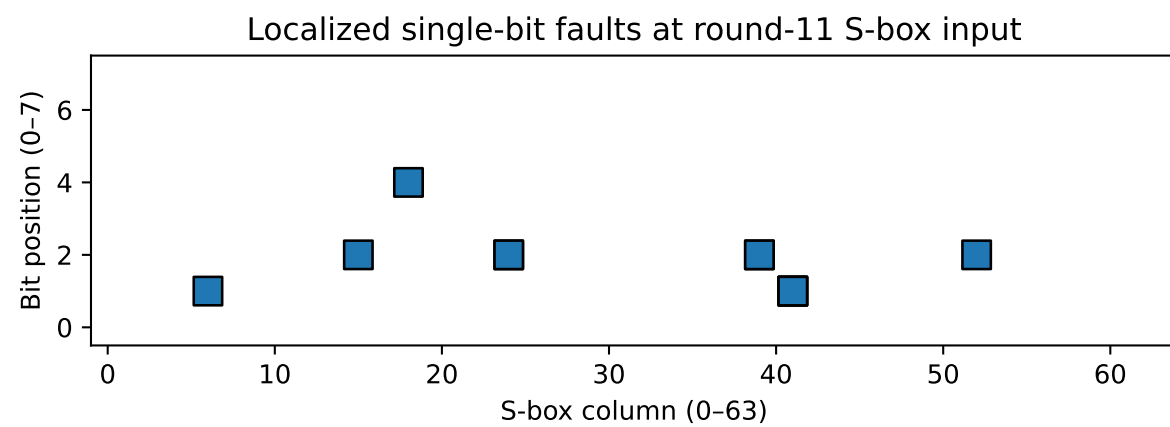
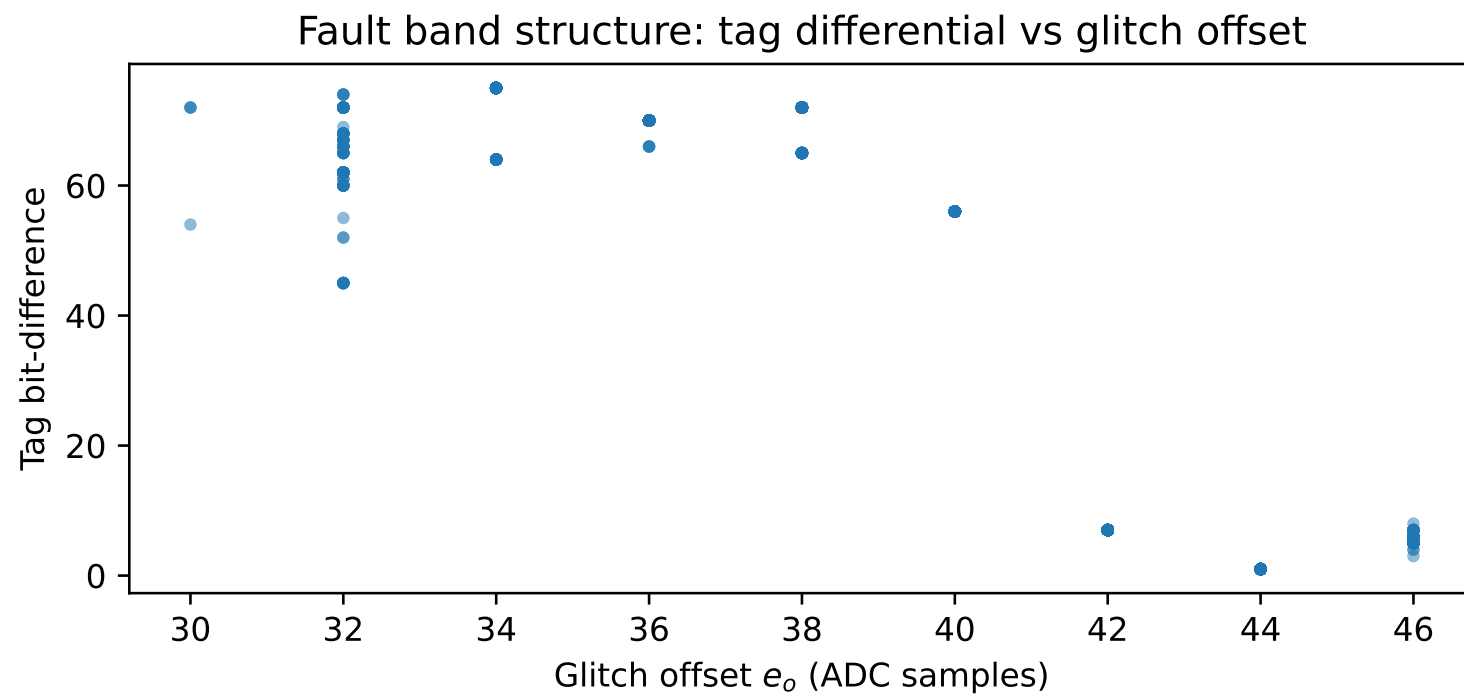
Closed-Loop Adaptive Key Recovery

Posterior-scoring chosen-nonce attack on one column, $M=1$: converges to the correct hypothesis in 32 queries, posterior 0.999, using only public information (power + chosen nonces). Per-column scope (2 of 128 bits) stated honestly; full-key assembly is the divide-and-conquer step above.

Masked-Core Comparison (d=1, 2-share)

Same pipeline against the threshold-implemented core (9,151 traces, random keys): round-1 S-box at chance for every architecture (CNN, MLP, centered-product MLP). First-order masking confirmed effective on hardware; the unmasked-vs-masked contrast isolates masking as the causal variable. Cost: $\approx 1.8\times$ throughput, $\approx 2.1\times$ area (LUTs).

Clock-Glitch Fault Injection



- Sweeping glitch offset $e_o \in [28, 50]$ reveals a stable fault band at $e_o=41$: large tag differentials (27-41 bits), two landing positions (sub-offset 0 / 3350).
- From 406 band hits over 153 distinct tags, 12 clean single-bit round-11 faults localized to exact (column, bit) — 10 distinct state columns — each verified by exact faulty-tag match under a simulated single-bit flip.
- Data-selected placement: each nonce faults exactly one column, reproducibly; DFA coverage bounded at one column/query. Nonce-set engineering (≈ 266 nonces, coupon-collector bound) or multi-glitch schedules restore full-column coverage.
- Burst glitches fault 400/400 queries but as multi-bit superpositions (30+ bits), not composable single-bit faults or round skips.

Five Leakage-Validation Gates

- Prior-driven edge scores. A per-trace log-likelihood “edge” reported +0.398 nats on data where every classifier sat at the majority floor — the edge measured the prior. Gate: held-out accuracy must beat the majority-class floor.
- Prior-only profiles + separating-nonce selection. A below-floor profile outputs the prior regardless of trace; the picker then inverts it $\Rightarrow$ posterior 0.993 on the wrong hypothesis (0% label match). Gate: profiles entering a closed loop must clear the floor gate.
- Public-nonce shortcut labels. At a fixed key, column labels are a function of two public nonce bits; a model reading nonce-load activity “cracks” vacuously. Gate: profile on random keys; verify against re-encryption.
- Key-reload register-bus artifact. A capture loop that rewrites the key before every trace embeds key-write bus activity: $|r| \approx 0.6$ at the load samples, up to 64/64 columns of fake key-bit leakage. Gate: the leak must survive a fixed-key capture (nonces varying).
- Structured-key / free-alignment false recovery. A ramp-key template attack reported 75.8% per-bit recovery; the reversed ramp scores identically and the bit-inverted key scores higher (76.6%) — impossible for a genuine read (must invert to $\approx 24\%$). Blind random-key capture: no signal. Gate: inversion-invariance + structure-preserving nulls + blind random-key control.

Attack-Surface Synthesis

- Unmasked + PA: S-box target trainable (42.7% key-rank top-1 vs 25% chance); fine-tuning closes the domain gap; per-column closed-loop recovery at posterior 0.999 in 32 queries; $\sqrt{M}$ averaging is the SNR lever. Gap: 64-column assembly (measurement budget).
- Unmasked + FI: verified single-bit round-11 faults with exact localization (12 cases, 10 columns); burst faults multi-bit; placement data-selected. Gap: intra-query column coverage.
- Masked core: factorable target suppressed to chance (9,151 traces); masking performs at first order on hardware.
- Artifacts: five trap classes with gates; the key-reload bus and structured-key/roll traps are, to our knowledge, unreported for LWC hardware evaluation.

Conclusions

- An oracle-verified CW305 platform delivers three hardware-verified result classes on the official unmasked NIST LWC ASCON-128 core.
- DL-PA recovers per-column round-1 S-box key bits; per-column closed-loop recovery converges at posterior 0.999 in 32 queries; the $\sqrt{M}$ lever lifts rank-1 from 0.25 to 0.94.
- Clock glitching yields verified single-bit round-11 faults at exact (column, bit) positions — with data-selected placement as a documented DFA constraint.
- Five reusable gates separate signal from artifact; two trap classes are platform-general and unreported to our knowledge.
- First-order masking (d=1) is confirmed effective on the same platform.
- Full 128-bit assembly is enabled and characterized as ongoing work.

Acknowledgments

PES University ECE (ISFCR); NewAE CW305/Husky platform. Poster accompanies the VLSID 2027 Student Research Forum submission.